CBCS SCHEME

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BPLCK105C/BPLCKC105

First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the basic principles of object oriented programming.	6	L1	CO1
	b.	Explain primitive data types used in java.	8	L1	CO1
	c.	What is type conversion and type casting? Explain with an example.	6	L1	CO1
		OR			
Q.2	a.	Explain two control statements used in Java with syntax and example.	6	L1	CO1
	b.	What is multidimensional array? Explain with an example program.	8	L1	CO1
	c.	Explain lexical issues used in java.	6	L1	CO1
		Module – 2			
Q.3	a.	Write a java program to show the use of the following operators: i) << ii) >> iii) >>>	6	L3	CO2
	b.	Explain logical operators with example program.	8	L1	CO2
	c.	Explain for each version of for loop with an example.	6	L3	CO2
		OR			
Q.4	a.	Write a note on: i) Break ii) Continue.	6	L1	CO2
	b.	Explain different types of if statement with example program.	8	L3	CO2
	c.	Write a program to demonstrate the use of ternary operator.	6	L3	CO2
		Module – 3			
Q.5	a.	Explain two ways of argument passing methods with example.	6	L1	CO3
	b.	What are constructors? Explain two different types of constructors with example program.	8	L3	CO3
	c.	Define method overloading. Explain constructor overloading with an example.	6	L2	CO3

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		OR			
Q.6	**a.**	Write a program to perform stack operations using proper class and methods.	**6**	**L3**	**CO3**
	b.	What is recursion? Explain with an example program.	**8**	**L1**	**CO3**
	c.	Explain: i) This keyword ii) The finalize( )	**6**	**L1**	**CO3**
		Module – 4			
Q.7	**a.**	What is Inheritance? Explain how a super class variable can reference a subclass variable.	**6**	**L2**	**CO3**
	b.	Explain method overriding with example program.	**8**	**L3**	**CO4**
	c.	Describe the two uses of super keyword with example.	**6**	**L3**	**CO4**
		OR			
Q.8	**a.**	Write a note on: i) Final class ii) Final method.	**6**	**L1**	**CO4**
	b.	Explain dynamic method dispatch with example.	**8**	**L3**	**CO4**
	c.	What is abstract class? Explain with an example program.	**6**	**L3**	**Co4**
		Module – 5			
Q.9	**a.**	Define package. Explain with an example program.	**6**	**L3**	**CO4**
	b.	What is an exception? Explain different mechanisms of handling the exceptions in java.	**8**	**L3**	**CO4**
	c.	Explain nested interfaces with example.	**6**	**L1**	**CO4**
		OR			
Q.10	**a.**	Describe different access specifiers used in java.	**6**	**L1**	**CO4**
	b.	Define Interface. Demonstrate the working of interface.	**8**	**L1**	**CO4**
	c.	Explain how one interface can be extended by another with an example program.	**6**	**L3**	**CO4**

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Subject Title : Basics of Java Programming Subject Code : BPLCK105C/205C

Question Number	Solution	Marks Allocated
1. a	Basic principles of OOP. i) Encapsulation ii) Inheritance iii) Polymorphism. Explanation of 3 principles.	1M 5M. 6M
1. b.	Primitive Data types List byte, short, int, long, char, float, double & boolean. Explanation of primitive data types.	1M 7M 8M
1. C.	Type conversion & type casting definition converting one data type into another data type is called type conversion. Process of forcibly converting one data type into another is called type casting. Explanation of type conversion & Type casting	2M. 4M. 6M
2. a.	Control Statements used in Java. 1) The if Statement 2) the for Loop Syntax: if (condition) Statement; for(initialization ; condition; iteration) Statement; Example for if & for control Statements.	1M. 1M 4M. 6M
2. b.	Multidimensional Array definition. Multidimensional arrays are arrays of arrays. Example of Multidimensional Array program. class TwoArray { public static void main(String args[])	1M. 7M. 8M

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Question Number	Solution	Marks Allocated

```
{  int twoD[][] = new int[4][5];
     int i, j, k = 0;
     for( i=0; i<4; i++)
       for (j=0; j<5; j++)
         twoD[i][j]= k;
        k++;
   }
  for( i=0; i<4; i++)
  for(j=0;j<5;j++)
   System.out.println( twoD[i][j]+" ");
    }
  }}
```

2.C. Lexical Issues used in Java. — 1M
White space, Identifiers, Literals, comments, separators & keywords.
Explanation of lexical issues. — 5M
6M

3. a — 2M

```
class ByteShift {
   public static void main (String args[])
{  byte a=64, b;
     int i;
     i = a<<2;
     b = (byte) (a<<2);
    System.out.println("original value of a:"+a);
    S.O.P("i & b :" +i+" " +b);
  }}
```

2M

```
class HexByte {
    public static void main(String args[])
  {  char hex[] = { '0','1','2','3','4','5',
       '6','7','8','9','a','b','c','d','e','f'.
    };
```

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Question Number	Solution	Marks Allocated
	byte b=(byte) 0xf1; System.out.println("b = 0x" +hex[(b>>4)& 0x0f]+hex[b& 0x0f]); }}	
	Unsigned Right Shift.	2M.
	Ex: int a = -1; a = a>>>24; 11111111 11111111 11111111 11111111 -1 >>>24 00000000 00000000 00000000 11111111	8M
3. b)	Logical operations & - Logical AND | - Logical OR ^ - Logical XOR ! - Logical NOT	2M

	Logical operators example program.
	class BoolLogic {
	public static void main (String args[])
	{ boolean a = true; boolean b = false;
	boolean c = a|b; boolean d = a^b;
	boolean e = a&b; boolean f = (a&!b)|
	(a^!b);
	boolean g = !a.
	System.out.println(" a = " + a);
	(" b = " + b);
	(" a|b = " + c);
	(" a^b = " + d);
	(" a&b = " + e);
	(" !a &b|a&!b = ", +f);
	(" !a = " + g);
	}}
		6M.
		8M.

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Question Number	Solution	Marks Allocated

3.C	General form of the for each version of the for loop.
	for (type itr-var : collection)
	statement block;
	Example.
	Class ForEach {
	Public Static void main (String args[])
	{ int nums[] = {1,2,3,4,5,6,7,8,9,10};
	int Sum = 0;
	for (int x : nums) { System.out.println("
	value is : " + x);
	Sum += x;
	} System.out.println ("Summation : " + Sum);
	}}
		1M
		5M
		6M

4.a.	i) break
	↳ It terminates a statement sequence in a switch.
	↳ exits a loop
	↳ civilized form of goto.
	Example. Class BreakLOOP{
	Public Static void main (String args[])
	{ for (int i=0; i<100; i++)
	{ if (i == 10) break;
	System.out.println ("i: " + i);
	}
	}}
	ii) Continue
	↳ A continue statement causes control to be transferred directly to the conditional expression that controls the loop.
		1M
		2M
		1M

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Question Number	Solution	Marks Allocated
	Example program class continue{ Public Static void main (String args[]) { for (int i=0; i<10; i++) { System.out.println (i+" "); if (i%2 == 0) continue; }}}	2M 6M
4b)	Types of if Statement if, if-else, Nested if, if-else-if ladder. Example Program.	1M 7M 8M
4c)	Ternary operator syntax Expression1 ? expression2 : expression3 Example program	1M 5M 6M
5 a)	Two ways of argument Passing methods (1) Pass by value (2) Pass by reference example program	1M 5M 6M
5. b)	Constructor definition Constructor is a block of codes similar to the method. Which is having same name as the class name. Two different types of constructors ① Default constructor ② Parameterized constructors. Example program.	1M 1M 6M 8M

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Question Number	Solution	Marks Allocated
5. c)	Method overloading definition Two or more methods within the same class that share the same name as long as their parameter declarations are different, the process is called method overloading. Example program for method overloading	1M 5M 6M

6. a)	Write a program for stack operations, push & POP.
	Example program.
	Class stack()
	{ int stck[] = new int[10];
	int tos;
	stack()
	{ tos = -1;
	}
	void push(int item)
	{ if(tos == 9)
	S.O.P("stack is full");
	else
	stck[++tos] = item;
	}
	void POP()
	{ if(tos < 0)
	S.O.P("stack underflow");
	return 0;
	else
	return stck[tos--];
	} }
		1M
		5M.
		6M

6. b)	Recursion Definition Recursion is the process of defining something in terms of itself. Recursion is the attribute that allows method to call itself. Example programs.	1M 6M 8M

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Question Number	Solution	Marks Allocated
6. c)	i) This Keyword can be used inside any method to refer to the current object.	1M
	Example.	2M
	```Box (double w, double h, double d) { this.width = w; this.height = h; this.depth = d; }```	
	ii) The finalize()	
	↳ By using finalization, Java defines specific actions that will occur when an object is just about to be reclaimed by garbage collector.	1M
	Syntax & Explanation	2M
	```protected void finalize() { // finalization code here }```	6M
7 a)	Inheritance Definition	1M
	↳ Inheritance is a concept that acquires the properties from one class to other class.	
	Example program for a superclass variable to reference a subclass object.	5M / 6M
7.b)	Method overriding definition	
	When a method in a subclass has the same name & type signature as a method in its superclass, then the method in a subclass is said to be override the method in superclass	2M
	Example program for method overriding.	6M / 8M

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Question Number	Solution	Marks Allocated
7. c)	Two uses of super keyword. ① A subclass can call a constructor defined by its superclass.. super(arg-list); Example program ② Superclass refers to the subclass of its own class in which it is used. Super.member Example program.	3M 3M 6M

8. a)	i) final class.
	if the class is defined as final, then it implicitly declare all of its methods as final.
	Example. final class A {
	//
	}
	class B extends A{ // Error ! can't Subclass A
	}
	ii) final Method.
	if the methods declared as final cannot be overridden.
	example: class A {
	final void meth() {
	} { System.out.println("This is a final method");
		1M
		2M
		1M
		2M
		6M

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Question Number	Solution	Marks Allocated
	class B extends A{ void meth() { S.O.P ("Illegal"); }}	
9. a)	Package Definition Packages are containers for classes that are used to keep the class name space compartmentalized.	1M
	General form of Package	1M
	Package pkg; Example program.	4M 6M
9. b)	Exception Definition A Java exception is an object that describes an exceptional condition that has occured in a piece of code.	1M
	Java Exception handling Mechanisms try, catch, throw, throws & finally. Example program.	1M 6M 8M
9. c)	Nested Interface An interface can be declared a member of a class or another interface. Such an interface is called a member interface or a nested interface	1M
	Example program	5M 6M

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Question Number	Solution	Marks Allocated
10. a)	Different Java acces Specifiers. private, public protected. Explanation of acesses Specifiers.	1M 5M 6M

10. b)	Definition of Interface.
	An interface specifies the behaviour of the Class by providing an abstract type.
	The general form of interface
	acces interface name{
	return-type method name1 (Parameter-list);
	return-type method name2(Parameter-list)
	type final-varname1 = value;
	type final-varname2 = value;
	//
	...
	return-type method-nameN(Parameter-list);
	} type-final-varNameN = value;
	Example program.
		1M
		1M
		5M
		8M

10. c)	interface can be extended using key word extends. Example program.	1M 5M. 6M.